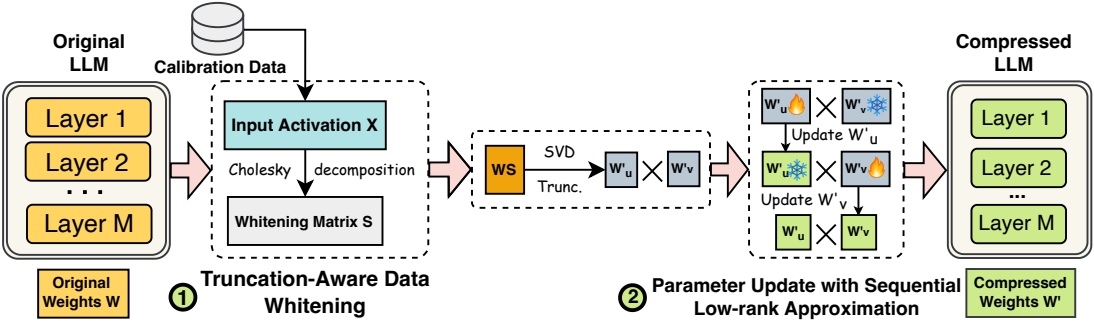

Original LLM
Layer 1
Layer 2
Layer M
Calibration Data
Input Activation X
Cholesky
decomposition
Whitening Matrix S
SVD
Trunc.
W's
W'u
W'v
Update W'u
Update W'v
Compressed LLM
Original Weights W
① Truncation-Aware Data Whitening
② Parameter Update with Sequential Low-rank Approximation
Compressed Weights W'